Determinants of the U.S. 10-Year Treasury Yield: A Regression Analysis from 2015 to 2025

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Abstract

This study investigates how macroeconomic and financial indicators influence the 10-Year Treasury Yield between 2015 to 2024. A multiple linear regression model was constructed using monthly aggregated data, with predictors including the federal funds rate, lagged inflation indicator, unemployment rate, Standard & Poor's 500 fluctuations, gold prices, and the U.S. dollar value index. Also, the special market movements brought by the COVID-19 period, especially its deep effect to unemployment rate, were considered as well. Diagnostic tests confirmed model validity, and Newey-West standard errors were used to correct for autocorrelation. Results show that the federal funds rate was the only consistently significant predictor, showing the strong influence of monetary policy on long-term yield rate. Interpretations of other variables' insignificance in our model suggest that yield movements were shaped not only by quantifiable indexes, but also a combination effect of investor expectations and policy signals, particularly during times of heightened uncertainty.

Introduction

The role played by the U.S. 10-Year Treasury Yield is indispensable in the U.S. economy and even the global economy. Basically, it refers to the interest rate earned by investors on a type of U.S. government-issued debt with a maturity of 10 year (Chen, 2024). It is also seen as the "risk-free" rate as it's backed by the full faith and credit of the U.S. government. The performance of the yield indicates numerous situations of the economy and financial markets, such as investors' confidence and expectation to economic growth. Not only to investors themselves, the 10-Year Treasury yield potentially affects other interest rates across the market, including mortgage rates, corporate bond yields, and bank loan rates (Zucchi, 2024). Therefore, for public to better understand what is going on in current economic situation and for investors to make wise decision, research and investigation on this indicator is essential, especially in discovering the factors which may influence this benchmark. For instance, if we predict the yield of US10Y (U.S. 10-Year) starts rising next few months, we may expect that companies will rush to issue corporate bonds prior to that, because the US10Y yield determines borrowing costs of issuing debt (Graham & Harvey, 1999).

Some of the influencing factors have already been proved by previous researchers, such as the prices of treasury notes (the price, or the market value of a bond, always moves in opposite direction to yield, because when the demand of treasury yield is high, the prices will naturally rise as well, and it further leads to lower yields (Zucchi, 2024)), worse stock market performance, lower interest rates (Federal Reserve policy), and lower inflation expectations, which we generally categorize them as "conventional factors." However, the main purpose of this research is to discover if there is anything else besides conventional ones that also potentially influences the yield. To do so, I assumed to bring 4 important indexes from financial,

economic, and political aspects into consideration: gold prices, foreign bond rates (yield), U.S. dollar value, and unemployment rate. These predictors are the most representative component of several aspects: precious metal (also considered as a safe-haven investment), global economic synchronization, domestic currency, and labor market. Ideally, indicators from these fields can effectively predict the US10Y trend.

From my observation, past studies on US10Y have primarily focused on only macroeconomic factors such as inflation and interest rate. However, if we can find underlying alternative indicators not only limited to macroeconomy, we may incorporate these factors to enhance the predictive power of a treasury trend prediction model.

Research Questions

Questions that this study aims to answer are:

1. What are the main drivers of change in the U.S. 10-Year Treasury Yield?
2. Do "unconventional" predictors significantly impact US10Y?

Statement of Purpose

If we can successfully find the answer to them, the forecasting model of US10Y will be improved, leading us a more accurate prediction to various economic and financial outcomes in the future.

Literature Review

Jiang et al. (2024) investigate the impact of macroeconomic variables – specifically, the savings rate, inflation rate, GDP growth rate, and interest rates – on the fluctuations of U.S. Treasury yields using data from 1986 to 2020. The study uses a linear regression model, extracting economic data from the past decade to determine the relationships among these

variables. Then, the model is applied additionally to European Union data to examine its usefulness across different economic systems. The findings in Jiang et al. (2024) indicate that U.S. interest rates and savings rates significantly influence Treasury yields, while inflation and GDP growth appear weaker in explanatory power. However, expanding the sample size to monthly data strengthened the statistical significance of inflation.

Similarly, Vissing-Jorgensen (2021) discusses the sharp spike in U.S. Treasury yields during the COVID-19 pandemic and the Federal Reserve's response to it in a qualitative framework. The study finds that the 10-year Treasury yield increased dramatically by 64 basis points between March 9 and March 18, 2020. Such scenario differs from the typical flight-to-safety behavior in financial markets. Then, the paper argues that this spike possibly due to a liquidity-driven selling by many financial institutions who have urgent cash needs, emphasizing liquidity shocks and central bank demand as major influences on Treasury markets. Therefore, this work uncover the important role that central bank in maintaining the stability of financial market during crises, suggesting that the yield curve responds not only to economic conditions but also in demand from investors.

Similarly yet differently, Hsing (2015) also examines the factors which might contribute the government bond yield, but in the market of Spain. The study explores the relationship between government debt and long-term interest rates in Spain. Based on the model and further analysis, Hsing (2015) finds that the Spanish government bond yield is positively correlated with the government debt-to-GDP ratio, short-term Treasury bill rate, expected inflation rate, U.S. 10-year government bond yield, and a debt crisis dummy variable; moreover, it is negatively associated with GDP growth and the expected nominal effective exchange rate. The findings

suggest that higher debt levels and inflation expectations raise borrowing costs, whereas economic growth and currency appreciation decrease them.

The three articles all focus on financial markets and macroeconomic factors, particularly their influence on government bond yields. Government Bond Yield in Spain (Hsing, 2015) and Study of Factors Influencing U.S. Treasury Yields (Jiang et al., 2024) analyze the effects of inflation, GDP growth, interest rates, and fiscal conditions on bond yields. Meanwhile, The Treasury Market in Spring 2020 and the Response of the Federal Reserve (Vissing-Jorgensen, 2021) focuses on the COVID-19-led spike in Treasury yields, investigating the Federal Reserve's role in stabilizing the market. Whereas the first two studies research on long-term macroeconomic relationships using regression models, Vissing-Jorgensen emphasizes the influence brought by liquidity shock and central bank interventions in a qualitative approach. Despite these differences on methodology, all three papers highlight the critical role of monetary and fiscal conditions in shaping Treasury markets.

Hsing (2015) and Jiang et al. (2024) both see interest rates and inflation as key determinants of bond yields at the beginning of the paper. Hsing specifically finds that a higher debt-to-GDP ratio and U.S. bond yields contribute to increased Spanish bond yields.However, GDP growth rate (of U.S.) does not show its significance to determine 10-year treasury yield in Jiang et.al 's case. Also, in contrast to Hsing (2015) and Jiang et al. (2024), Vissing-Jorgensen (2021) attributes the Treasury yield fluctuation during COVID-19 to liquidity-driven selling instead of causation of macroeconomics indexes.

Overall, in the prediction of future 10-year Treasury Yields, all three studies have considered key macroeconomic indicators such as interest rates, monetary expectations, and inflation. However, few have directly combined and compared the significance of traditional

predictors listed above and other seemingly less obvious predictors within one single model, particularly using post-2015 U.S. data. This research seeks to fill that gap by introducing a broader set of predictors, including the traditional predictors such as federal funds rate, inflation, unemployment rate, stock market fluctuations, as well as uncommon predictors such as gold prices, foreign bond yields, and the U.S. dollar index.

Methodology

Participants

The population in this study refers to the broader set of macroeconomic conditions under which the 10-Year Treasury Yield responds to key financial and economic indicators. The data used in this study consisted of monthly observations collected from January 2015 to December 2024. Each data point represents macroeconomic and financial conditions observed in a given month.

Data Collection

All datasets are processed, cleaned, and merged using Python. The data used in this study all comes from reliable sources, retrieved from FRED (Board of Governors of the Federal Reserve System, 2024; U.S. Bureau of Labor Statistics, 2024), Yahoo Finance (Yahoo Finance, 2024), and Wall Street Journal (Wall Street Journal, 2024).

Dependent variable:

- Treasury Yield (%) from FRED (Board of Governors of the Federal Reserve System, 2024)

Independent variables:

- Stock Market Index (in index value): S&P 500 index (Yahoo Finance, 2024)

- Interest Rate (Federal Funds Rate) (%) from FRED (Board of Governors of the Federal Reserve System, 2024)
- Inflation Rate: calculated based on Consumer Price Index (U.S. Bureau of Labor Statistics, 2024)
- Precious Metals Prices: Gold (in USD / ounce) from Yahoo Finance (Yahoo Finance, 2024)
- Foreign Bond Yields: China 10Y (%) from WSJ (Wall Street Journal, 2024)
- U.S. Dollar Index (in index value) from Yahoo Finance (Yahoo Finance, 2024)
- Unemployment Rate (%) from FRED (U.S. Bureau of Labor Statistics, 2024)

Among those datasets, US10Y Yield (response), S&P 500, Gold, China 10Y, and US Dollar Index are reported daily, whereas the others are reported monthly. Hence, I select the end-of-month "Close Price" for daily-based data to generalize the overall behavior of the month.

For data who follow a monthly frequency (i.e., Federal Funds Rate, CPI, Unemployment Rate), the available monthly report is published on the first day of next month; therefore, to ensure timeframe alignment, the collected data for Federal Funds Rate, CPI, and Unemployment rate starts from 2015-02-01 to 2025-01-01, which represents the performance over January 2015 to December 2024.

As a result, the final merged dataset includes monthly data starting from 2015-01-31 to 2024-12-31, covering 120 months. The output after running a "missing values" test shows that the merged dataset is complete and contains zero missing value, which means the current dataset is ready to use for further statistical analysis.

Variables

The response variable is the U.S. 10-Year Treasury Yield from January 1, 2015 to December 31, 2024, which is the yield over the recent 10 complete years. The sample selected is a monthly data over the 10-year span. I am planning to aggregate daily yields into monthly value by setting the yield on the end-of-month day (e.g., using data on 2015.12.31 to represent the overall performance in December). This approach, undoubtedly, will bring drawbacks that we lose a plenty of observations compared to daily data; however, the advantage of doing so is not only a matter of convenience, but we also eliminate the insignificant day-to-day volatility, which can enhance our data and graph to be neat and smooth. In response, we also aggregate data of predictor variables who are originally recorded daily to fulfil data alignment. Thus, we would, ideally, have 120 rows of data (before cleaning) representing 120 months in total.

The explanatory variables included are 7 in total: 1) Stock prices index – the index I will use to represent U.S. stock performance is S&P 500 (Standard & Poor's 500, which tracks the performance of the leading 500 companies in U.S. stock market); 2) Interest rate (U.S.) – the index I will use to represent overall interest rate is Federal Funds Rate, a crucial indicator of monetary policy and also has significant influence on other interest rates); 3) Inflation rate – I select monthly change of Consumer Price Index (CPI), which reveals the change of money paid by consumers over time, to indicate the inflation; 4) Gold prices; 5) Foreign bond rates – I select China 10-Year Government Bond, counterpart of US10Y, to represent this factor; 6) U.S. dollar index which gives us the value of U.S. dollar; 7) U.S. unemployment rate.

It is worth noting that the inflation rate was lagged by two periods (i.e., months) to account for the time delay in how market participants incorporate inflation information into their expectations of long-term interest rates – therefore, we are basically using the inflation rate 2 months ago as the predictor. Similarly, the S&P 500 index was transformed into a first-difference

variable to focus on short-term fluctuations in stock market performance rather than long-term price levels. This is due to the idea that bond yields may be more sensitive to recent market fluctuation rather than simply absolute equity values.

Statistical Modeling and Analysis Plan

The model includes a range of predictors assumed to be significant from both economic theory and assumptions: the federal funds rate (FEDFUNDS_RATE), U.S. dollar index (USD), gold prices (GOLD), unemployment rate (UNRATE), the first difference of the S&P 500 index (SP500_diff), the inflation rate lagged by two months, and the 10-Year Chinese Government Bond Yield (CN10Y).

Additionally, a dummy variable indicating the COVID-19 pandemic period (March 2020 to June 2021) is included to account for structural shifts during this unusual market phase. Motivated by patterns observed in exploratory scatterplots, an interaction term between unemployment rate and the COVID dummy is introduced to examine whether the relationship between unemployment and long-term interest rates differed during the pandemic.

$$Y_t = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \beta_8 Z_1 + \beta_9 X_4 Z_1 + \varepsilon$$

- Y_t - US10Y in time t (%)
- $\beta_0, \beta_1, \ldots, \beta_7$ – coefficients for intercept and each predictor respectively
- X_1 – FEDFUNDS_RATE (%)
- X_2 – USD (%)
- X_3 – GOLD (%)
- X_4 – UNRATE (%)
- X_5 – SP500_diff
- X_6 – inflation_lag2 (%)

- X_7 – CN10Y (%)
- Z_1 – COVID (=1 for pandemic period [March 2020 to June 2021]; = 0 otherwise)
- ε – error term

Analysis Plan

Several diagnostic tests will be performed to test the validity of the regression models. An F-test is used to evaluate the overall performance of the model, and individual predictor coefficients are tested using t-tests. Given the time-series nature of the data, a Durbin-Watson test will be conducted to detect whether autocorrelation exists. If significant evidence of autocorrelation does take place, the model will be adjusted using Newey-West heteroskedasticity and autocorrelation-consistent (HAC) standard errors. Multicollinearity will be checked through variance inflation factors (VIFs) and correlation matrix. Linearity will be checked through scatterplots, and the presence of heteroskedasticity will be explored using residual plot.

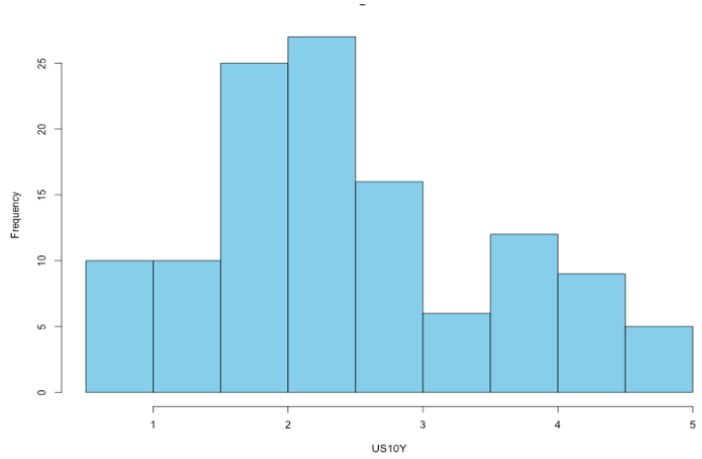
Results

This section states the results of a multiple linear regression analysis investigating the influence of macroeconomic and financial predictors on the 10-Year U.S. Treasury Yield (US10Y). The analysis primarily focuses on identifying which predictors are statistically significant, based on robust estimation using Newey-West standard errors. In addition to the baseline model, an extended version was also estimated to test for potential interaction effects between unemployment and the COVID-period. Results are reported in terms of overall model fit, the statistical significance of individual variables, and relevant diagnostic tests to ensure the validity of model.

Descriptive Statistics

The response variable of interest is the US 10-Year Treasury Yield (US10Y), a widely followed indicator of economic expectations in the United States. Over the sample period (from 2015-01-31 to 2024-12-31), the average US10Y yield was approximately 2.472%, with a standard deviation of 1.074. The yield ranged from a minimum of 0.55% to a maximum of 4.88%. The histogram in **Figure 1** shows a moderately skewed distribution, with a cluster of observations around the 1.5 – 2.5% range.

Figure 1 Histogram of US10Y



Among the explanatory variables, the federal funds rate (FEDFUNDS_RATE) is important as it shows a strong positive correlation with the response variable. Its average value is 1.818% over the sampled period. Another key variable worth to be specified, the unemployment rate (UNRATE) shows sharp peaks around 2020 due to the COVID-19 pandemic. This allows us to evaluate structural shifts during the pandemic and makes

things easier if we incorporate an interaction term between the UNRATE and COVID in the future. The inflation rate was lagged by two period to take the time it takes to reflect market performance into account, instead of static price levels.

Additionally, the S&P 500 stock index was transformed into a first-order difference variable to *appropriately* reflect the effect of fluctuation of stock prices to the yield, rather than the static price levels.

Table 1 Descriptive Statistics of Predictors

Variable	Min	Mean	Max	SD
Fed Funds	0.050	1.818	5.330	1.881
UNRATE	3.40	4.67	14.80	1.731
Gold Price	1060	1605	2738	397.2
USD	89.13	98.12	112.12	4.907
China 10Y	2.010	3.056	4.031	0.436
Inflation (lagged)	-0.792	0.254	1.295	0.2776
SP500 Diff.	-398.5	32.22	374	156.98

Scatterplots and correlation analysis were used to investigate relationship between US10Y and other predictor variables. Each scatterplot includes both a red linear regression line and a blue curve to capture the actual relationships between each predictor and the 10-Year Treasury Yield. The gray band around the red line represents a 95% confidence interval for the linear fit. The scatterplot in **Figure 2** reveals strong positive correlations between US10Y and Fed Funds Rate. However, in **Figure 3** and **Figure 4**, the scatterplots show that little correlation existed or is distorted by outliers between US10Y and inflation rate as well as S&P 500 index difference, and our correlation matrix (**Table 2**) shows similar result. Moreover, CN10Y, both

through scatterplot (**Figure 5)** and correlation matrix (**Table 2)**, shows little or no linear relationship with US10Y.

To add control variable, a COVID dummy was introduced. The boxplot (**Figure 6**) comparing US10Y during COVID vs. non-COVID periods suggest an obvious decline in yields during the pandemic period. through scatterplot (**Figure 5)** and correlation matrix (**Table 2)**, shows little or no linear relationship with US10Y.

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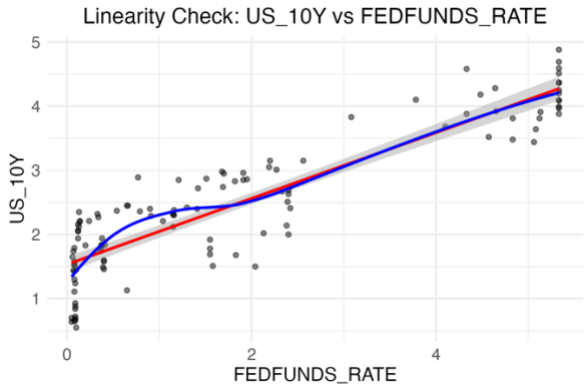

Figure 2 Scatterplot of Fed Funds Rate vs. US10Y

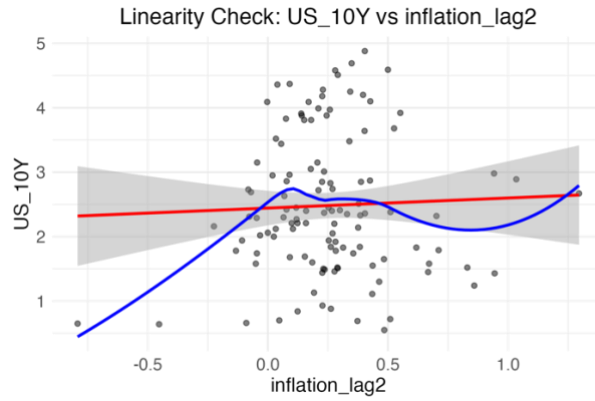

Figure 3 Scatterplot of Inflation Rate (lagged) vs. US10Y

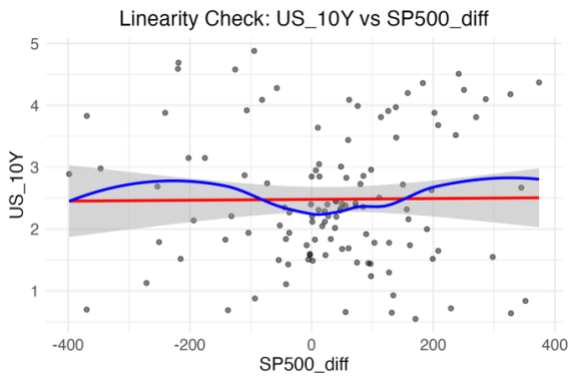

Figure 4 Scatterplot of SP500_diff vs. US10Y

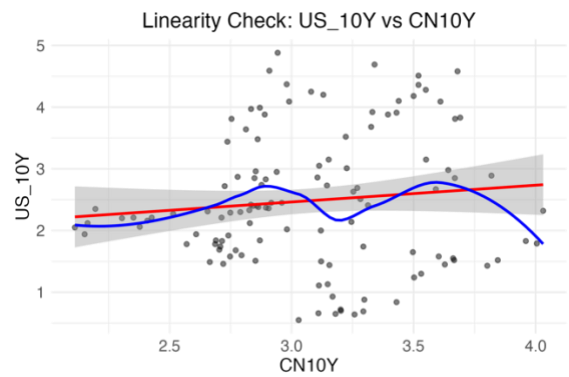

Figure 5 Scatterplot of CN10Y vs. US10Y

Table 2 Correlation Matrix

	US_10Y	FEDFUNDS	CN10Y	GOLD	inflation_lag2	SP500_diff	UNRATE	USD
US_10Y	1							
FEDFUNDS_RATE	0.896	1						
CN10Y	0.106	0.157	1					
GOLD	0.421	0.584	0.623	1				
inflation_lag2	0.04	-0.037	0.426	0.237	1			
SP500_diff	0.01	0.124	-0.04	0.157	-0.094	1		
UNRATE	-0.568	-0.441	-0.04	-0.003	-0.303	0.038	1	
USD	0.704	0.693	0.212	0.445	0.015	-0.087	-0.253	1

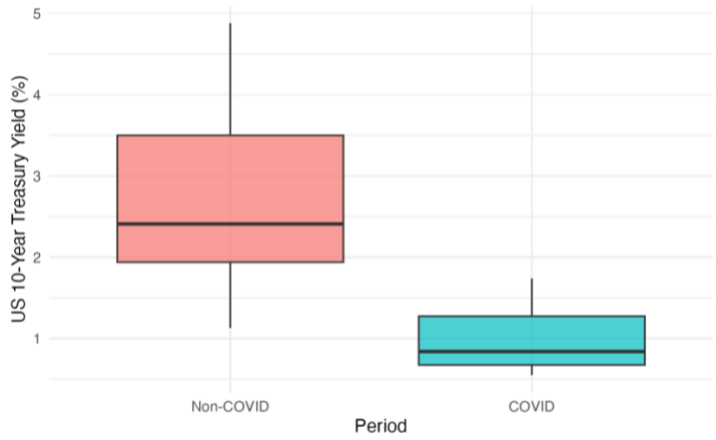

Figure 6 Side-by-side Boxplot of US10Y & COVID

Inferential Statistics

To investigate which macroeconomic and financial factors influence the 10-Year U.S. Treasury Yield (US10Y), a multiple linear regression model was constructed with the following predictors: the federal funds rate (FEDFUNDS), USD index (USD), the first difference of the S&P 500 index (SP500_diff), a two-period lag of the inflation rate (Inflation_Lag2), the unemployment rate (UNRATE), a dummy variable for the COVID-19 period (COVID), and an interaction term between UNRATE and COVID. We dropped CN10Y after considering its non-linearity and little correlation to response variable exhibited in the scatterplot and correlation matrix.

Table 3 shows the model summary. In the initial model, FEDFUNDS_RATE, USD, UNRATE, COVID, and UNRATE*COVID all showed statistical significance (individual t-test with p-value < 0.05 for all) in determining the trend of treasury yield. Specifically, In **Table 3,** sr^2 represents each predictor's individual explanatory power after controlling other predictors in the model. In alignment with the t-test results, FEDFUNDS_RATE showed the highest value of

sr^2, which further indicates its significance. An F-test indicated that the overall model was statistically significant (F(8, 109) = 103.9, p < .001).

Table 3 Regression results using US_10Y as the criterion

Predictor	*b*	*b* 95% CI [LL, UL]	sr^2	sr^2 95% CI [LL, UL]	Fit
(Intercept)	-2.29*	[-4.41, -0.16]			
FEDFUNDS_RATE	0.50**	[0.42, 0.58]	.16	[.09, .23]	
USD	0.03**	[0.01, 0.06]	.01	[-.00, .02]	
GOLD	-0.00	[-0.00, 0.00]	.00	[-.00, .01]	
UNRATE	0.24**	[0.07, 0.41]	.01	[-.00, .02]	
covid	1.47**	[0.44, 2.50]	.01	[-.00, .02]	
SP500_diff	-0.00	[-0.00, 0.00]	.00	[-.00, .01]	
inflation_lag2	0.24	[-0.08, 0.56]	.00	[-.00, .01]	
UNRATE:covid	-0.35**	[-0.53, -0.17]	.02	[-.00, .03]	
					R^2 = .884** 95% CI[.83,.90]

However, once after the Durbin-Watson test was conducted and revealed extremely strong autocorrelation (p-value ≈ 0.00), the initial linear regression model was no longer reliable. Therefore, the final regression model was based on Newey-West standard errors which fixes basic autocorrelation problem. After running the final version, the results (**Table 4)** indicated that the FEDFUNDS_RATE (federal funds rate) was the only variable that remained statistically significant at the 5% level (SE = 0.1568 t = 3.1826, p < 0.002). This finding reinforces the dominant role of monetary policy in shaping long-term treasury yield expectation.

Although a few variables were significant in original linear regression model, its effect was no longer statistically significant once autocorrelation was corrected by using Newey-West standard errors. These variables, while theoretically important, might reflect a combination of factors which we will explain in detail later. The interaction term between unemployment and COVID was initially significant under original linear regression model, but lost significance under HAC standard errors, suggesting that its effect may be partly explained by autocorrelated residuals.

Table 4 Regression coefficients with Newey-West standard errors for US10Y model

Predictor	Estimate	Std. Error	t value	p value	Sig.
(Intercept)	-2.2865	8.0489	-0.28	.7769	
FEDFUNDS_RATE	0.4989	0.1568	3.18	.0019	**
USD	0.0334	0.0634	0.53	.5994	
GOLD	-0.0003	0.0008	-0.31	.7575	
UNRATE	0.2399	0.2884	0.83	.4073	
covid	1.4693	1.5905	0.92	.3576	
SP500_diff	-0.0004	0.0002	-1.76	.0814	
inflation_lag2	0.2380	0.2030	1.18	.2421	
UNRATE × covid	-0.3515	0.3187	-1.10	.2724	

Model Diagnostics

To assess the validity and quality of the multiple linear regression model, several diagnostic checks were performed. The model exhibited a Multiple R-squared of 0.8841 (**Table 3)** and an Adjusted R-squared of 0.8756. This suggests that approximately 88% of the variation in US10Y can be explained by the included predictors.

Given that macro-financial time series data often show time-based correlation, a Durbin-Watson test was initially conducted, revealing significant positive autocorrelation in the residuals. To account for this, HAC standard errors were estimated using the Newey-West method. This correction affected the statistical significance of several predictors, most notably

the unemployment rate and its interaction with the COVID period, which were significant under original model (p-value ≈ 0.00024) but lost significance under robust standard errors.

Multicollinearity was assessed using variance inflation factors (VIFs), and variables all exhibit eligible VIF values, with all VIFs falling well below the common threshold of 4.

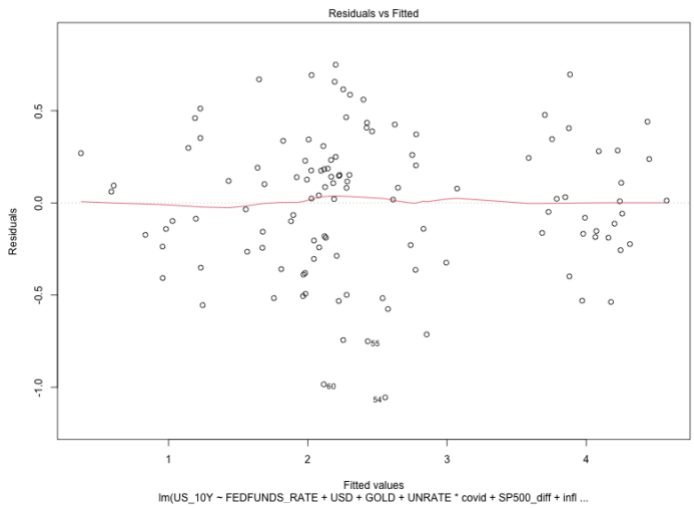

Figure 7 Residual Plot

Residual plots (**Figure 7)** from the model did not exhibit strong signs of heteroscedasticity or non-linearity, supporting the feasibility of the final model. While only the FEDFUNDS_RATE remained statistically significant after robust adjustment ($p < .01$), the overall model maintained relatively strong explanatory power and satisfied key diagnostic tests.

Conclusion

This section illustrates the conclusion we draw from the results of our statistical models and analysis, acknowledging the insufficiency in this study and providing suggestions for future research among the same area.

Interpretation of Results

The statistical result above revealed that the federal funds rate was the only statistically significant predictor of the 10-Year U.S. Treasury Yield after we applied Newey-West robust standard errors. The positive coefficient suggests that increases in the federal funds rate are associated with increases in 10-Year Treasury Yield, reinforcing the dominant influence of monetary policy on bond markets during the sample period. While previous research and Federal Reserve Bank suggest that the federal funds rate does not always move closely with the long-term yield rates (Martin, 2017), the strong statistical relationship observed in this model may reflect several historical events taken place during the sample period (2015-2024). We experienced period of zero lower bound (ZLB), which is an action made by the Fed to boost spending by lowering the short-term rate (Bernanke, 2017), from March 2020 to March 2022 in response to the economic recession brought by pandemic, and meanwhile market participants, during this time period, may have placed increased weight on short-term rate, such as federal funds rate, when making long-term interest expectations. According to previous study done by Swanson & Williams (2014) on similar situation taken place in 2008, long-term Treasury yields appear to reflect market expectations about the future trend of the federal funds rate. This may explain why, in our model, the federal funds rate appears to be the only consistently significant predictor, while alternative variables lose explanatory power once robust corrections are applied.

For other predictors, although a few macroeconomic indicators such as previous inflation rate and fluctuation of stock index are frequently discussed in other studies as key drivers of long-term interest rates, their lack of significance in this model may be due to either overlapping effects with the federal funds rate, insufficient sample period, or the limitations of a linear model in capturing their more complex or indirect influence.

The absence of significance of inflation is especially noteworthy given its essential role in affecting the yield rate from theoretical perspective. This result might be attributed to the complexity of economic policy. To be specific, it is important to recognize that market participants often form expectations about future economic conditions (including yield rates) not by reacting to inflation rates directly, but rather by interpreting monetary policy signals provided by the Federal Reserve. In many cases, yields adjust before inflation really takes place, based on public interpretation of the Fed's announcements and anticipated actions. For example, when the Fed announces a potential rate cut, the yield may increase the next day – that is not due to the "rate-cut" is the direct cause, but because investors expect that such action will eventually lead to higher inflation, so they react prior to the inflation really happens. However, our model cannot observe the whole process of public expectation. It simply identifies that a change in the federal funds rate correlates with a change in the 10-Year Treasury Yield, without capturing the underlying psychology of investors. In this situation, the insignificance of inflation in our model does not imply that it is irrelevant to yield. Instead, its effects may be embedded in the policies or anticipated by the market beforehand. Therefore, the absence of statistical significance should not be mistaken for economic unimportance.

In contrast, for those predictors who were initially significant but lost explanatory power after handling autocorrelation – such as U.S. dollar index and the interaction term between unemployment and the COVID period – this shift suggests that the original significance may have been overestimated due to autocorrelation in the residuals, which violates the condition of a linear regression assumption. While these variables may still be meaningful in economic terms, their effects are not significant under strict statistical criteria.

Implication

The findings of this study provides implications for understanding the behavior of long-term treasury yield. The strong association between the federal funds rate and the 10-Year Treasury Yield suggests that investors likely use short-term policy signals to make long-term expectations. Meanwhile, it reinforces the Federal Reserve's decisions on short-term policies largely influence long-term investment decisions. In practice, the results suggest that investors, analysts, and policymakers should closely monitor short-term rate changes, especially in special time periods which economy is under uncertainty. Furthermore, even though the purpose of this study is to find determinants of long-term treasury yield out of seven candidates, this model could be further refined for forecasting purpose as well, such as adding a lagged response variable term to increase the overall predicting power of the model.

Limitations

While the model explains a considerable portion of the variance in the 10-Year Treasury Yield, several limitations must be acknowledged. First, the use of monthly data may fail to capture short-term market dynamics or high-frequency responses to news and policies, which are the essential features of financial world. Second, the model assumes a linear relationship between the predictors and the yield, which may not adequately capture underlying nonlinear relationships for particular predictors. Third, although the inclusion of a dummy referring pandemic period and interaction term captures some structural changes, more nuanced shifts in expectations and investor behavior due to other events may not be fully captured. Finally, not only the relatively short post-2015 sample period, which only contains 120 data entries, but also an unusual pandemic era included within this time period, limits the model's ability to generalize findings to broader macroeconomic contexts. Ultimately, financial markets are shaped not only by quantifiable economic indicators but also by rapidly evolving expectations, sentiment shifts,

and unforeseen political events – many of which lie beyond the scope of statistical modeling. From a single policy statement to a sudden global crisis, the real-world factors of long-term yields are often too complex to be fully captured by purely quantitative predictors. This underscores the inherent limitations of any empirical model and the need to interpret results from a broader understanding of financial behavior.

Recommendation

Future research could improve the study by using models other than a linear framework, such as models with time-varying parameters. This may help capture the dynamics and time-based trait of most economic indicators. Moreover, sample size could be expanded to include a longer historical period; alternatively, researchers can increase the time frequency from monthly to weekly. In addition, future research could incorporate investor sentiment indices or measures of forward-looking behaviors to better take public's expectation process into account. Finally, as this research is conducted on U.S. bond market and may not be suitable for foreign bond markets, it would also be valuable to test whether similar results hold in other countries' government bond markets.

Summary

In summary, this study examined the relationship between several key macroeconomic/financial indicators and the 10-Year U.S. Treasury Yield from 2015 to 2024. It finds that among all predictors, the federal funds rate remains the most consistent and useful driver of U.S. 10-Year Treasury Yield in recent macroeconomic conditions. This result illustrates the dominant role of monetary policy in shaping long-term interest rates even during a period of economic uncertainty and policy intervention. Also, within the context, these results reveal the complexity of modeling financial behavior in real-world expectation-driven market environment.

While the statistical analysis models specific relationships, it cannot fully capture investor's psychological behavior, policy signals, and global uncertainty. This study enhances our understanding of bond market but also reminds us of the limitations in modeling in real-world cases.

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